

Shashi Bhushan Vijay

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SUMMARY

Results-driven Integrated M.Tech Computer Science Engineering student and patent holder specializing in machine learning pipelines, distributed systems, and backend engineering. Demonstrated ability to build production-grade systems achieving measurable outcomes - including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE & PROJECTS

OncoFollow - Full-Stack Oncology Follow-Up Platform

Apr 2026 – Jun 2026

Project

Repo

- Built OncoFollow, a full-stack oncology follow-up platform that unified patient reporting, clinician review, admin oversight, document uploads, and longitudinal timelines into a single care workflow.
- Engineered a rule-based triage system covering 20 symptom categories and 4 clinical scoring dimensions (severity, duration, frequency, trend) to classify cases into 4 urgency tiers (routine/soon/urgent/emergency).
- Implemented a safety-first escalation layer with red-flag detection and override logic; delivered 99% sensitivity for emergency-case interception in synthetic evaluation.
- Designed explainable decision support by grounding recommendations in NCCN/ASCO guidance plus recent patient records, producing citation-backed outputs for clinical traceability and trust.
- Developed secure AI integration through server-side API routing and role-specific prompt controls, enabling structured, non-diagnostic support across patient and clinician experiences at production-style architecture scale.

FLEX-DCA AI Platform - AI-Powered Debt Collection Allocation & Compliance System

Jan–Feb 2026

Project

Repo

- Built a full-stack debt recovery platform using FastAPI, React, and SQLite to ingest cases, run predictions, auto-allocate DCAs, and track operational KPIs via REST APIs.
- Developed and integrated 3 CatBoost models (recovery probability, aging risk, closure speed) with financial feature engineering for real-time decision support.
- Implemented a RAG-based legal copilot using FAISS + SentenceTransformers + Groq LLaMA 3.1, enabling policy-grounded answers for FDCPA, Regulation F, and RBI compliance queries.
- Created an operations dashboard with KPI monitoring and case workflows, improving visibility into recovered/open cases and allocation outcomes.
- Structured backend into modular services (ML, allocation, RAG, dashboard, ingestion) to support scalable, production-style architecture for enterprise debt operations.

AutoML Assistant

Jan 2026

Project

Repo

- Built an end-to-end AutoML platform using Streamlit + Python to automate data upload, preprocessing, analysis, model training, and optimization workflows.
- Developed a modular ML backend supporting regression, classification, and time-series forecasting with algorithms like XGBoost, Random Forest, ARIMA/SARIMAX, and optional LSTM.
- Integrated LLM-powered model recommendation (RAG) using LangChain, Groq (Llama 3.1), and FAISS to suggest suitable models based on dataset statistics and ML best practices.
- Implemented SHAP-based explainability (feature importance, beeswarm, waterfall, dependence plots) to make model predictions interpretable for end users.
- Exposed trained models through a FastAPI inference API (`/predict`, `/predict/csv`) with persistent workspace/model management for reusable deployment-ready ML pipelines.

- Developed an end-to-end indoor localization system using Wi-Fi RSSI fingerprinting and Bluetooth integration, achieving approx. 1.6 m positioning accuracy (MAE: 1.468 m, R²: 0.978) in complex indoor environments.
- Built an ESP32-based autonomous Wi-Fi sniffer mounted on a robotic car, automating radio-map generation and reducing manual site-survey effort while collecting 4,277 RSSI records across 209 access points.
- Designed and optimized a Dynamic Stacking ensemble model combining KNN, Random Forest, and XGBoost with a Gradient Boosting meta-learner, improving localization performance by approx. 43% (RMSE: 3.124 to 1.787) over conventional stacking approaches.
- Engineered a real-time data pipeline using MQTT, Raspberry Pi, and Python, transforming raw Wi-Fi scans into machine-learning-ready datasets and enabling location predictions within the system’s real-time constraints.
- Implemented an A*-based indoor navigation module using NetworkX, generating optimal routes from predicted user locations to destinations and enabling dynamic path recalculation for real-time indoor guidance.

EDUCATION

Vellore Institute of Technology, Vellore	Expected April 2027
Integrated M.Tech in Computer Science Engineering	GPA: 7.0/10
Relevant Coursework: Machine Learning, Distributed Systems, Data Structures & Algorithms, Database Management, Computer Networks, Operating Systems, Cloud Computing	
SM Public School, Haridwar	2020–21
Junior College (Class XII)	91.2%
SM Public School, Haridwar	2018–19
High School (Class X)	87%

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, SQL

Backend & Systems: FastAPI, REST APIs, Microservices, MQTT, Docker, CI/CD

Databases: PostgreSQL, SQLite | Tools & Cloud: Git, AWS, GCP, Streamlit

ML & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, LSTM, SHAP, RAG, FAISS, LangChain, Groq

Methodologies: Agile, Scrum, DevOps, Data Engineering, Big Data

LEADERSHIP & EXTRACURRICULARS

- Sponsorship Coordinator, ISTE Chapter – VIT Vellore
- Secured **Rs. 1,00,000+ in corporate sponsorships** for flagship hackathon (Horizon) by pitching to 100+ founders and startups; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Built a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate

Designing Seamless Notification Service – View Certificate